Manmit Samal

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PROFESSIONAL SUMMARY

AI/ML student developer with hands-on experience building and deploying end-to-end machine learning pipelines using Python, FastAPI, Docker, and MongoDB. Skilled in supervised learning, data preprocessing, CNN architectures, and LLM-based integrations. Passionate about building practical, production-ready AI solutions that go beyond notebooks — from containerized backends to live deployed web apps.

EDUCATION

B.Tech in Electronics and Computer Science (EXCS) *2024 – 2028*
Vidyalankar Institute of Technology (VIT) Mumbai, India

- **Academic Performance:** 9.5 (Sem 1), 9.58 (Sem 2), 9.57 (Sem 3), Sem 4 (Ongoing)
- **Certification:** Data Science with Generative AI — *PW Skills (In Progress)*
  - **Modules Completed:** Python, Machine Learning, Statistics
  - **Upcoming:** Deep Learning, NLP, Generative AI, MLOps

TECHNICAL SKILLS

Languages	Python, C++, Dart
AI / ML	NumPy, Pandas, Seaborn, Matplotlib, Scikit-learn, TensorFlow, Keras, Regression, Classification, Clustering, Data Preprocessing, Model Training, API-based LLM Integration
Development	Flutter, FastAPI, MongoDB, Supabase, Postman
Tools	Git, GitHub, Docker, VS Code, Jupyter Notebook, Google Colab
Others	Figma, Notion, Streamlit

PROJECTS

Spam Detection *End-to-End ML Pipeline · FastAPI · Docker · MongoDB*

- Built a full spam detection ML pipeline integrating MongoDB for data ingestion, SMOTE for class imbalance handling, and Neuro-MF hyperparameter tuning.
- Dockerized FastAPI backend serving real-time SMS/Email predictions via an HTML UI, enabling cloud-ready deployment with a single `docker-compose` command.
- Designed a modular `src/pipeline` architecture separating ingestion, transformation, training, and serving — following enterprise MLOps conventions.

Customer Personality Segmentation *End-to-End ML Pipeline · XGBoost · Docker · AWS*

- Built an end-to-end ML prediction system orchestrating K-Means clustering and XGBoost classification to segment customers with **96% accuracy**.
- Developed a production-ready FastAPI backend to serve predictions and persisted user transactions in a NoSQL MongoDB database.
- Containerized the full application stack with Docker for environment consistency and cloud deployment readiness on AWS.

Dog vs Cat Image Classifier *Keras · CNN · Streamlit · Live Demo*

- Built and deployed a CNN image classifier distinguishing cats from dogs (**81% test acc**, 75% val acc); identified dataset size as the primary bottleneck and mitigated via augmentation — live at dog-cat-classifier-cnn.streamlit.app.
- Applied Data Augmentation (`ImageDataGenerator`) and Batch Normalization to reduce overfitting; documented lessons learned in `PROJECT_REVIEW.md`.
- Compared Keras and PyTorch training loops across MNIST and CIFAR-10 to understand framework-level differences in CNN workflows.